

Ayush Maheshwari

Senior Solutions Architect, NVIDIA

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Experience

NVIDIA

Senior Solutions Architect, NVAITC

Gurugram

Oct 2024 – Present

- Foundation Models: Building multilingual and domain-specific foundation models for Indian languages and scientific domains.
- Research Collaborations: Collaborating with premier academic institutions and research organizations.

Vizzhy Inc

Lead Research Scientist, Language Model Team

Bengaluru

Sep 2023 – Sep 2024

- Led a team of 5 people to build Indic large language models from scratch.
- Training the model on large accelerator cluster.

Adobe Research

Research Intern, Document Cloud Team

Bengaluru

May – Aug 2021

NCETIS, IIT Bombay

Project Engineer

Mumbai

May 2016 – Dec 2018

Water & Climate Lab, IIT Gandhinagar

Junior Research Fellow

Gandhinagar

May 2015 – Apr 2016

Tata Consultancy Services

Systems Engineer

Mumbai

Oct 2011 – June 2013

Education

Indian Institute of Technology Bombay

Ph.D. in Computer Science and Engineering

Defence: July 2024; 'Impactful Research Award 2023' by IIT Bombay

Mumbai, India

Jan 2019 – Aug 2023

Thesis: Knowledge Integration in NLP Models using Constraint Ingestion and Generation

Advisor: Prof. Ganesh Ramakrishnan

Tata Institute of Social Sciences

M.Sc.

Mumbai, India

2013 – 2015

University College of Engineering

B.Tech. (CSE)

Rajasthan Technical University Kota, India

2007 – 2011

Open-Source contributions

- SPEAR** : Data-Programming python library to create training data.
- UDAAN** : Post-editing workbench to quickly create publishable translations.

[GitHub](#)

[Website](#)

Research Interests

- Develop large language models for mid and low-resource languages.
- Develop machine learning based methods that generalize to new domains with limited resources.
- Lexically constrained neural machine translation, data programming, semi-supervised learning

Awards, Services & Talks

- My PhD thesis work was awarded **Impactful Research Award 2023** by IIT Bombay
- Best Demo Paper Award at CODS-COMAD 2023
- Awarded prestigious Ekal Fellowship 2020
- Reviewer: ARR 22–24, EACL, EMNLP, ACL (21–23)
- Fully covered grant for attending EMNLP 2022, Abu Dhabi
- Co-Founder at non-profit [Temples of India](#) - A web and app platform listing > 700K temples across Indian subcontinent.
- Key Member of [Udaan Project](#) - A project to translate education textbooks in Indian languages.

Computer Skills

Programming Languages: Python, C, C++, Javascript/jQuery, Bash, SQL

Libraries: PyTorch, DeepSpeed, HuggingFace, NumPy, scikit-learn

Talks

- Delivered Masterclass on **Building data-efficient large language models** [NLTF NASSCOM](#), 2024
- Invited Tutorial on **Data Efficient Machine Learning for end-to-end Educational Content Creation** [International Conference on Educational Data Mining](#), 2023
- Talk on **SPEAR: Data Programming using Semi-Supervision** [\[Talk\]](#) [DECILE Workshop](#), 2022

Publications

1. AQBENCH: Benchmarking Neural Surrogates for Air Quality Forecasting *NeurIPS (ED Track)*, 2026
Siddharthan D, Bedi S, Parmar P, **Maheshwari A**, Kota S, Krishnan N
2. IndicParam: Benchmark to evaluate LLMs on low-resource Indic languages *EMNLP (Main)*, 2026 [↗](#)
Maheshwari, A, Sharma K, Patel V, Maheshwari A *ACL (Main)*, 2025
3. LexGen: Domain-aware Multilingual Lexicon Generation [↗](#)
Maheshwari, A, Singh A, Karthika NJ, Bhat K, Jyothi P, Ramakrishnan G
4. ARISE: Iterative Rule Induction and Synthetic Data Generation for Text Classification *Findings of NAACL*, 2025
Yashwanth M, Singh V, **Maheshwari, A**, Krishna A, Ramakrishnan G [↗](#)

5. DictDis: Dictionary Constrained Disambiguation for Improved NMT [↗](#)
Findings of EMNLP, 2024
Maheshwari, A, Jyothi P, Ramakrishnan G
6. FAIR: Filtering of Automatically Induced Rules [↗](#)
EACL 2024
Bajpai, D.J, **Maheshwari, A**, Hanawal, M. and Ramakrishnan, G
7. Samayik: A Benchmark and Dataset for English-Sanskrit Translation [↗](#)
LREC-COLING 2024
Maheshwari, A, Gupta, A, Krishna, A, Singh, A.K, Ramakrishnan, G, Gourishetty, A.K. and Singla, J
8. Adaptive mixing of auxiliary losses in supervised learning [↗](#)
AAAI, 2023 (Oral Presentation)
Sivasubramanian D, **Maheshwari, A**, Prathosh AP, Shenoy P, Ramakrishnan G
9. UDAAN-Machine Learning based Post-Editing tool for Document Translation [↗](#)
CODS-COMAD Demonstrations, 2023 (Best Demo Paper Award)
Maheshwari, A, R. Ajay, Subramanian V, Ramakrishnan G
10. Eigen: Expert-Informed Joint Learning Aggregation for High-Fidelity Information Extraction from Document Images [↗](#)
Machine Learning for Health (ML4H) at NeurIPS 2023
Singh, A, Venkatapathy, **Maheshwari, A**, Narayan, P, Shetty, D.P. and Ramakrishnan, G
11. Learning to Robustly Aggregate Labeling Functions for Semi-supervised Data Programming [↗](#)
Findings of ACL, 2022
Maheshwari, A, Killamsetty K, Ramakrishnan G, Iyer R, Danilevsky M, Popa L
12. A Benchmark and Dataset for Post-OCR text correction in Sanskrit [↗](#)
Findings of EMNLP, 2022
Maheshwari, A, Singh, N, Krishna, A. and Ramakrishnan G,
13. Data Programming using Semi-Supervision and Subset Selection [↗](#)
Findings of ACL, 2021
Maheshwari, A, Chatterjee O, Killamsetty K, Iyer R, Ramakrishnan G
14. Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification [↗](#)
EACL 2021
Maheshwari A, Chatterjee S, Ramakrishnan, G. Jagaralpudi S.
15. OCR On-the-Go: Robust End-to-end Systems for Reading License Plates & Street Signs [↗](#)
International Conference on Document Analysis and Recognition, 2019
Saluja, R, **Maheshwari, A**, Ramakrishnan, G, Chaudhuri, P. and Carman, M
16. DynGAN: Generative Adversarial Networks for Dynamic Network Embedding [↗](#)
Graph Representation Learning Workshop at NeurIPS 2019
Maheshwari, A, Goyal, A, Hanawal, M.K. and Ramakrishnan, G
17. Semi-supervised data programming in python [↗](#)
EMNLP 2022 - System Demonstrations
Abhishek, G, Ingole, H, Laturia, P, Dorna, V, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R,

18. Rule Augmented Unsupervised Constituency Parsing [↗](#)
Findings of ACL, 2021
Sahay, A, Nasery, A, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R
19. Representation learning on graphs by integrating content and structure information [↗](#)
11th International Conference on Communication Systems & Networks (COMSNETS)
Maheshwari, A, Goyal, A, Kumar, A, Hanawal, M.K. and Ramakrishnan, G
20. Enhancing Low-Resource NMT with a Multilingual Encoder and Knowledge Distillation: A Case Study [↗](#)
LoResMT Workshop at ACL 2024
Roy, A, Ray, P, **Maheshwari, A**, Sarkar, S. and Goyal, P
21. Unsupervised learning of explainable parse trees for improved generalisation [↗](#)
IJCNN 2021
Sahay, A, **Maheshwari, A**, Kumar, R, Ramakrishnan, G, Hanawal, M.K. and Arya, K
22. Tale of tails using rule augmented sequence labeling for event extraction [↗](#)
Star-AI Workshop at AAAI 2019
Maheshwari, A, Patel, H, Rathod, N, Kumar, R, Ramakrishnan, G. and Bhattacharyya, P
23. Entity Resolution and Location Disambiguation in the Ancient Hindu Temples Domain using Web Data [↗](#)
NAACL 2018 - System Demonstrations
Maheshwari, A, Kumar, V, Ramakrishnan, G. and Nath, J.S